

PROBLEM STATEMENT #27 | Smart Cities, Transport & Logistics

Warehouse Inventory & Pallet Location Tracker

1. Problem Context & Overview

An automated warehouse management system tracking 3D shelf pallet positions, enforcing shelf maximum weight limits, and logging stock movements via barcode/RFID scanning.

Target Stakeholders / Actors: *Warehouse Operator, Logistics Supervisor*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.

Sample Acceptance Criteria: Pass: Pallet assigned to valid rack; Fail: Placement permitted on rack where total weight exceeds rated structural limit.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The warehouse item lookup service must locate any SKU pallet coordinate across 100,000 bins in under 100 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.